

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/01

Paper 1 Multiple Choice

19 September 2014

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **20** printed pages.

[Turn over

- 1** A certain organelle in a eukaryotic cell was isolated and analysed. It was found that the organelle contains proteins, nucleotides and phospholipids.

Which organelle(s) could it possibly be?

- 1 nucleus
- 2 lysosome
- 3 ribosome
- 4 mitochondrion

- A** 1 only
- B** 2 and 3
- C** 1 and 4
- D** 1, 3 and 4

- 2** Which statement is true for phospholipids, but not for protein?

- A** It has hydrophilic and hydrophobic components.
- B** It is synthesized from non-identical sub-units.
- C** It can form a barrier to water soluble molecules.
- D** It is found in cell membranes.

- 3** A carbohydrate test procedure was carried out on the urine of an infant who was nourished with a synthetic milk formula. A sugar was found to be in the urine. This compound was tested positive for Benedict's test but was tested negative when tested with Clinistix test (a test used by diabetics). Acid hydrolysis of the urine sample did not increase the amount of reducing sugar measured.

Which of the following compound is most likely to be present in this infant's urine?

- A** glucose
- B** fructose
- C** maltose
- D** sucrose

- 4 Equal quantities of a 5% starch solution were added to equal quantities of different solutions.

After 5 minutes at room temperature, the solutions were tested with iodine in potassium iodide solution and the depth of colour was recorded using a colorimeter (with an arbitrary scale of 1 to 5, 5 being the darkest blue).

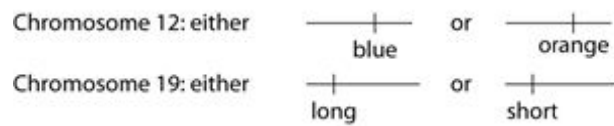
The results are given in the table.

solution	colorimeter reading
1 saliva and dilute HCl	4
2 saliva and dilute NaHCO ₃	1
3 dilute HCl only	4
4 saliva at 100°C	5
5 saliva at 20°C	3

This experiment suggests that saliva contains a biologically active substance that

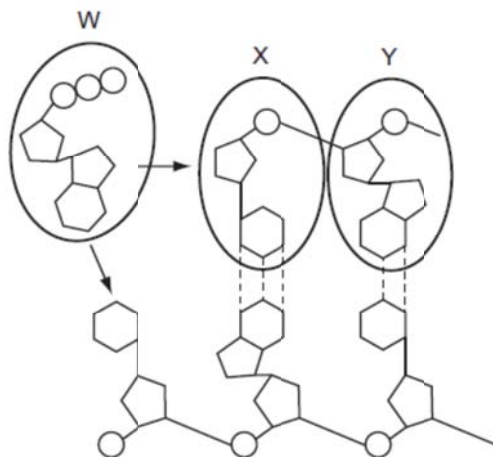
- A hydrolyses starch in acid conditions
 - B is denatured by hydrochloric acid
 - C is promoted in its action by high temperatures
 - D may be activated by hydrogen carbonate ions
- 5 When cytoplasm from a cell that is undergoing mitosis is injected into another cell that is in interphase, the cell receiving the cytoplasm
- A terminates in interphase
 - B enters mitosis
 - C enters apoptosis
 - D loses its p53 protein

- 6 Researchers have just discovered a new species of an insect. This organism is diploid. This insect may have blue or orange wings as the consequence of one of its genes on chromosome 12, and it may have either long or short wings as the result of a second gene on chromosome 19, as shown below.



If a female of this species has one chromosome 12 with a blue gene and another chromosome 12 with an orange gene, and has both number 19 chromosomes with short genes, she will produce which of the following egg types?

- A only blue, short gene eggs
 B one-half blue, short and one-half orange, short gene eggs
 C three-fourths blue, long and one-fourth orange, short gene eggs
 D three-fourths blue, short and one-fourth orange, short gene eggs
- 7 The diagram shows the synthesis of a polynucleotide. Molecule W is a nucleotide triphosphate.



Which statements are correct?

- 1 The base in W could be the purine, adenine
 - 2 The base in Y is the purine, guanine
 - 3 The base in X is the pyrimidine, cytosine
 - 4 The base in X could be the pyrimidine, uracil
- A 1 and 3
 B 2 and 3
 C 2 and 4
 D all of the above

- 8 The table below shows a list of characteristics displayed by mutant strains of *E.coli* during DNA replication and the possible reasons.

No.	Characteristics of mutant <i>E.coli</i>	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is not complete	DNA ligase activity is missing
2	Parental DNA molecule does not unzip	DNA helicase is hyperactive
3	Synthesis of complementary strands are slow	DNA polymerase keeps dissociating from the DNA and has to re-associate
4	No initiation of replication occurs	A-T rich region at origin of replication deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

- A 1 and 3
 B 2 and 3
 C 1, 3 and 4
 D all of the above
- 9 Which statements about the genetic code are correct?
- 1 The genetic code is degenerate.
 - 2 There is only one codon for the amino acid methionine.
 - 3 Codons act as 'stop' and 'start' signals during transcription and translation.
 - 4 Prokaryotes generally use the same genetic code as eukaryotes.
- A 1 and 2
 B 1, 2 and 3
 C 1, 2 and 4
 D all of the above
- 10 5' – CAU – 3' is a codon in mRNA that specifies the amino acid histidine (His) in position 58 in the α chain of haemoglobin. What is the corresponding anti-codon in tRNA?
- A 5' – CAU – 3'
 B 3' – AUG – 5'
 C 3' – GUA – 5'
 D 5' – GUA – 3'

- 11** The petals of the morning glory flower are purple when the flower is in bud but become blue as the flower opens. The colour change is caused by an increase in pH of the vacuoles of the petal cells.

A mutation in a gene coding for a type of hydrogen ion (H^+) pump in the vacuole membranes of the petal cells results in flowers that are purple when open.

The difference in DNA sequence of the gene coding for the ion pump in plants with blue or purple flowers is shown below.

plant with blue flowers	-T-T-A-A-T-C-C-T-G-A-G-A-T-T-T-
plant with purple flowers	-T-T-A-A-T-C-C-T-G-C-T-G-A-G-A-T-T-T-

Which statements explain the purple colour of the flowers of the mutant plant?

- 1 A frameshift mutation does not alter the sequence of amino acids in the ion pump.
- 2 The primary structure and the shape of the ion pump are altered.
- 3 The activity of the ion pump is altered by changes in its shape.
- 4 The mutant ion pump cannot pump H^+ ions into the vacuoles of the petal cells

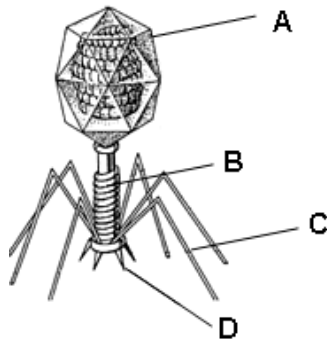
- A** 1 and 2
B 2 and 3
C 3 and 4
D 2, 3 and 4

- 12** Which of the following statement(s) about viruses is / are true?

- 1 They encode genes for synthesising their own ATP.
- 2 They are single cell organisms.
- 3 They can have a genome made of DNA.
- 4 They package ribosomes into their virion.
- 5 They can have a single stranded or double-stranded RNA genome.
- 6 They can have a membrane-like envelope.

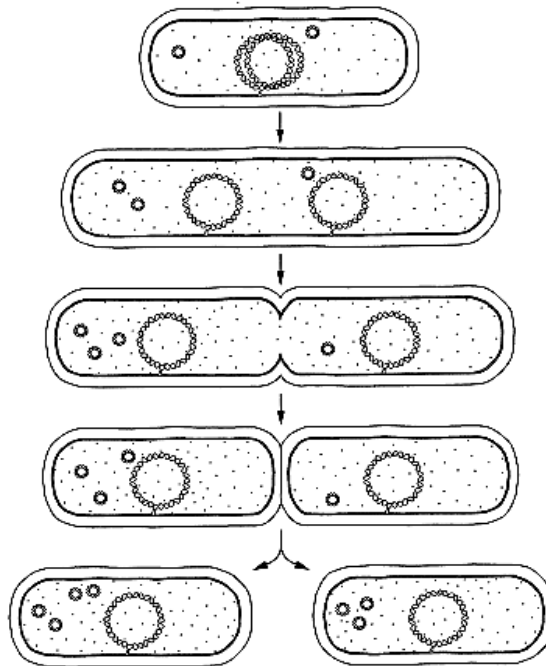
- A** 5 and 6
B 3, 5 and 6
C 1, 3, 5 and 6
D none of the above

13 The diagram shows a bacteriophage.



Which structure plays the most important role in host recognition?

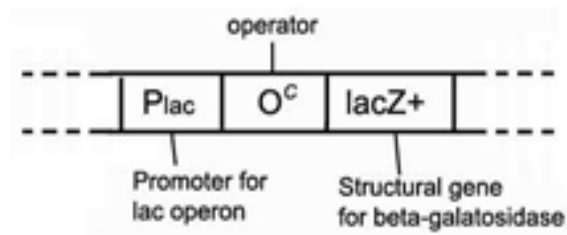
14 The diagram shows the reproduction of a bacterial cell.



Which of the following can be concluded from the figure?

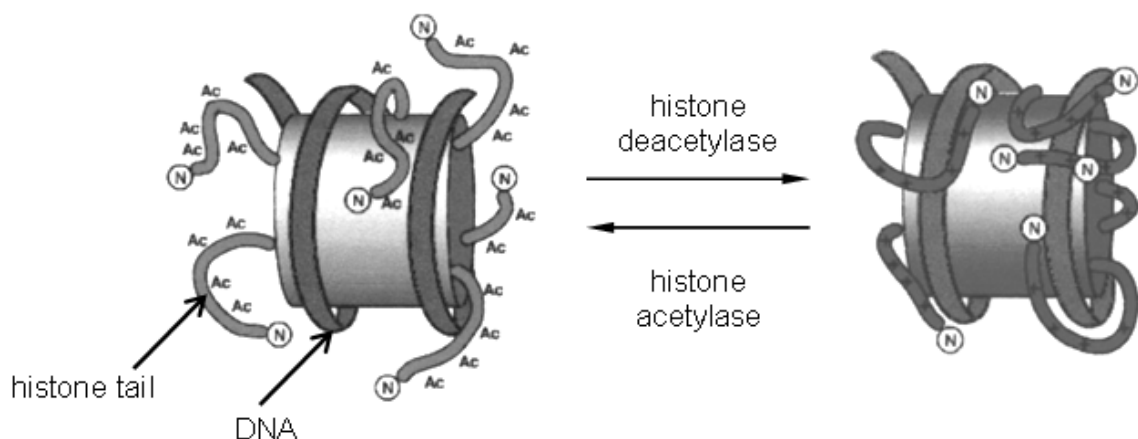
- A All DNA molecules are able to control their own replication.
- B This type of nuclear division results in the formation of genetically identical offspring.
- C Some DNA can be excised to form smaller DNA molecules.
- D This type of reproduction divides the cell into two daughter cells by forming a cell plate.

- 15 The *lac* operon of an *E. coli* strain carries the P_{lac} O^c *lacZ*⁺ DNA sequence as shown below. In this sequence, the *lacZ* allele is normal (*lacZ*⁺) but the operator (O^c) is mutated. The *lac* repressor is unable to bind to the O^c mutant.



How would the expression of the *lacZ* gene be affected in this strain of cells?

- A There will be normal regulation of lactose metabolism.
 - B There will be continuous expression of beta-galactosidase regardless of glucose levels.
 - C The inability to synthesis the *lacZ* gene product, beta-galactosidase.
 - D The *lac* operon is inducible to synthesize beta-galactosidase.
- 16 Which of the following is the best description of control elements?
- A They comprise of distal control elements and promoter.
 - B They are coding region of DNA sequence that regulates transcription.
 - C They are non-coding regions which bind to proteins called RNA polymerase.
 - D They comprise of enhancer and silencer.
- 17 The diagram below shows how acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression.



- A Helicase action is enhanced by acetylation.
- B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
- C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
- D RNA polymerase works better by binding with acetyl groups.

18 How is translation controlled in eukaryotes?

- A** By differential removal of introns enabling a gene to code for more than one protein.
- B** By activation of the protein by folding or cleavage after it is formed.
- C** By the production of RNA from the non-coding strand of the DNA.
- D** By protein factors that bind to specific sequences in the mRNA.

19 Which of the following conditions must be present for cancer formation?

	number of alleles mutated for each proto-oncogene	number of alleles mutated for each tumour-suppressor gene
A	1	1
B	2	1
C	1	2
D	2	2

20 Which of the following statement(s) about cancer is / are true?

- 1 Cancer is a result of decreased cell death due to the mutation of a proto-oncogene.
- 2 Individuals who inherit one inactive copy of tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies.
- 3 The lack of anchorage dependence during cell cycle control and failure to undergo differentiation are characteristics of cancerous cell.
- 4 Cancerous cells invade other tissues by metastasis, killing normal cells in the process.

- A** 2 and 4
- B** 1, 2 and 4
- C** 2, 3 and 4
- D** all of the above

21 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

- A one gene with a pair of co-dominant alleles
- B one gene with multiple alleles
- C sex linkage of the allele for grey coat colour
- D two genes, each with a dominant and recessive allele

22 Two genes A and B are linked together as shown below.

$$\begin{array}{c} \underline{A \quad b} \\ a \quad B \end{array}$$

If the genes are far enough apart such that crossing over between the alleles occurs occasionally, which statement is true of the gametes?

- A All of the gametes will be Ab and aB.
- B There will be 25% Ab, 25% aB, 25% ab and 25% AB.
- C There will be approximately equal numbers of Ab and ab gametes.
- D The number of Ab gametes will be greater than the number of ab gametes.

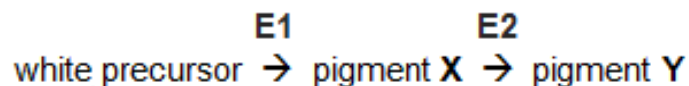
- 23 In the fruit fly, *Drosophila melanogaster*, four genes whose recessive alleles code for black body (B / b), curved wings (C / c), purple eyes (P / p) and vestigial wings (V / v) are linked on chromosome 2.

The table shows some distances between the gene loci, as determined by breeding experiments.

gene loci	distance between loci / map units
B / b and P / p	6.0
B / b and V / v	18.5
P / p and V / v	12.5
P / p and C / c	21.0
V / v and C / c	8.5

What is the correct sequence of the loci on chromosome 2?

- A B / b P / p V / v C / c
 B C / c V / v B / b P / p
 C P / p V / v C / c B / b
 D V / v B / b P / p C / c
- 24 The foxglove plant has three different coloured flowers - purple, pink and white. Coloured pigment is synthesised enzymatically from a white precursor as shown below:



Two genetically identical plants with white flowers were crossed. This produced an F₁ generation with 2170 white-flowered, 530 purple-flowered and 180 pink-flowered plants.

The alleles are represented by the following symbols:

- R**: no production of pigment **r**: production of pigment
B: purple pigment **b**: pink pigment

Which statement correctly describes the relationship between the two genes?

- A Dominant epistasis in which the epistatic allele is **B**.
 B Recessive epistasis in which the epistatic allele is **b**.
 C Dominant epistasis in which the epistatic allele is **R**.
 D Recessive epistasis in which the epistatic allele is **r**.

- 25 Which one of the following describes conditions in a photosynthesizing cell, exposed to high light intensity and low carbon dioxide concentration?

	concentration of CO ₂ acceptor	concentration of ATP	concentration of GP
A	high	high	high
B	high	high	low
C	low	high	high
D	low	low	high

- 26 Which of the following statements regarding cyclic and non-cyclic photophosphorylation is true?

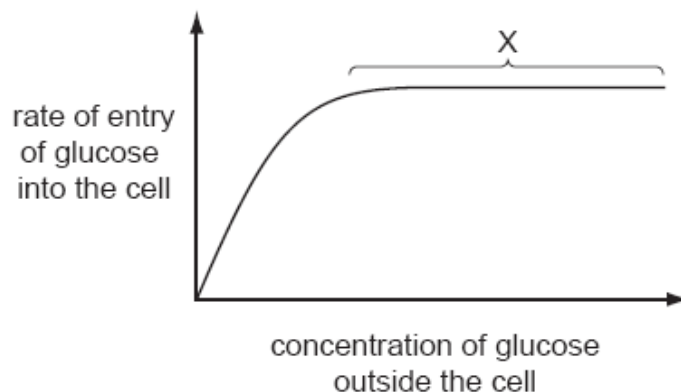
- A Cyclic photophosphorylation only requires light at 680nm to function optimally while non-cyclic photophosphorylation requires light at both 680 nm and 700 nm to function optimally.
- B Cyclic photophosphorylation is dependent on products generated through non-cyclic photophosphorylation.
- C Only non-cyclic photophosphorylation utilizes electrochemical gradient generated to drive ATP production.
- D Only cyclic photophosphorylation can function in the absence of photosystem II.

- 27 Which one of the following substances, when added, would directly result in a rapid decline in ATP production in glycolysis?

- 1 A chemical that would bind to NAD⁺ irreversibly and induces its reduction to NADH.
- 2 An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.
- 3 A chemical that creates an anaerobic environment by reducing the oxygen in the environment
- 4 A reagent that binds to the active site of ATP synthase permanently

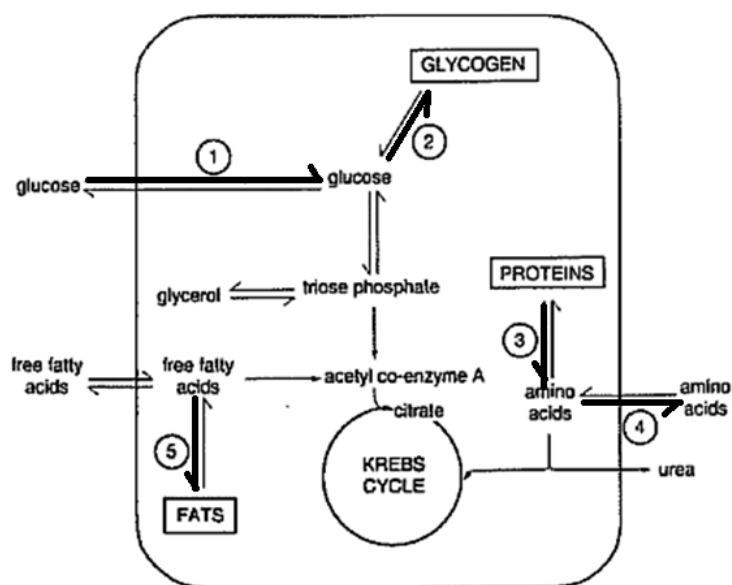
- A 1 and 2
- B 3 and 4
- C 1, 2 and 4
- D all of the above

- 28 The graph shows how the rate of entry of glucose into a cell changes as the concentration of glucose outside the cell changes.



What is the cause of the plateau at X?

- A All the carrier proteins are saturated with glucose.
 - B The carrier proteins are denatured and no longer able to function.
 - C The cell has used up its supply of ATP.
 - D The concentrations of glucose inside and outside the cell are equal.
- 29 The diagram shows some biochemical pathways in a liver cell. Some of the points where hormones affect the pathways are labelled 1 to 5.



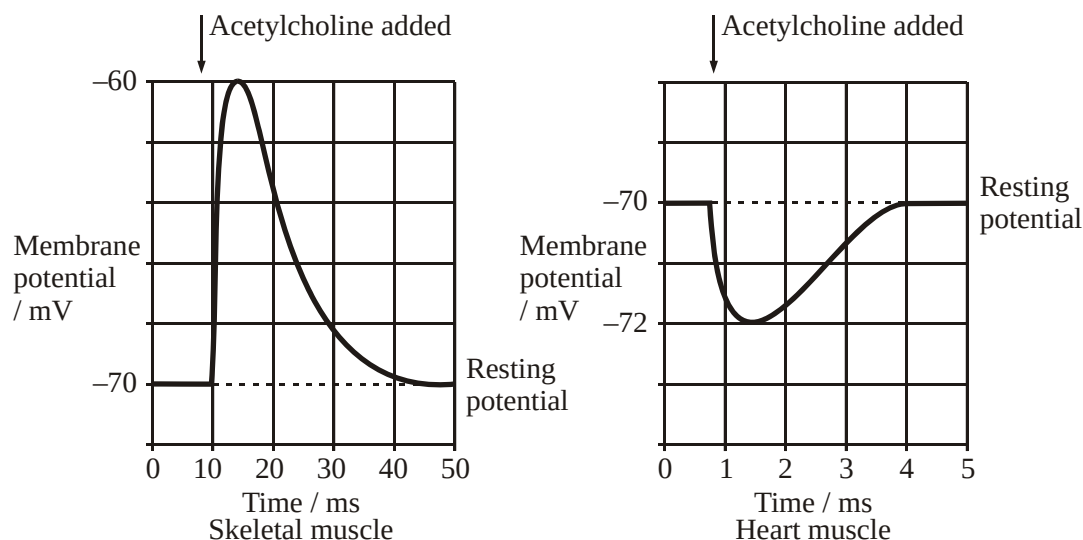
At which points would the hormone insulin accelerate the pathways in the directions indicated by the bold arrows?

- A 1, 2 and 3
- B 1, 2 and 5
- C 1, 3 and 4
- D 2, 3 and 4

30 Which of the following is not a consequence of increased cellular levels of cAMP?

- A Activation of a kinase cascade
- B Activation of adenylyl cyclase
- C Increased phosphorylation of glycogen phosphorylase
- D Inhibition of glycogen synthesis

31 The graphs below show the effect of adding acetylcholine to skeletal muscle and heart muscle.



Which of the following statements are true of the results above?

- 1 Depolarisation occurred when acetylcholine was added to skeletal muscle but hyperpolarisation occurred when acetylcholine was added to heart muscle.
- 2 Sodium channels opened when acetylcholine was added to skeletal muscle but potassium channels opened when acetylcholine was added to heart muscle.
- 3 Acetylcholine causes contraction of the skeletal muscle and causes a decrease in heart muscle contraction.
- 4 Binding of acetylcholine leads to formation of IPSP in skeletal muscle and EPSP in heart muscle.

- A 1 and 2
- B 2 and 4
- C 1, 2 and 3
- D all of the above

32 Which of the following statement(s) about genetic drift is / are true?

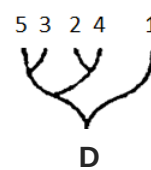
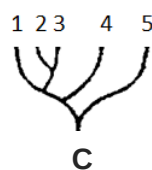
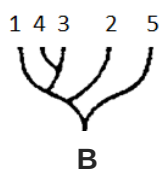
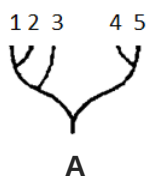
- 1 Genetic drift could reduce genetic variation
- 2 Genetic drift has more impact to a population if the beneficial alleles are at a very high frequency compared to those beneficial alleles at a very low frequency
- 3 Genetic drift is driven by the Founder effect and Bottleneck effect
- 4 Genetic drift occurs only in small populations

- A** 3 only
B 1 and 3
C 1, 2 and 3
D 1, 3 and 4

33 The cytochrome oxidase b gene of 5 species (**1-5**) of lizards on an island chain was studied. The DNA sequence of this gene in each of the 5 species was sequenced and analysed. The results showing the number of differences in the DNA sequences in the cytochrome b gene between the 5 species of lizards are presented below.

	1	2	3	4	5
1					
2	8				
3	10	4			
4	20	19	18		
5	26	25	24	20	

Based on the evidence in the table, deduce the probable evolutionary relationships of the 5 species of lizards from the tree diagrams below.



- 34 Researchers studying the genetic variation of marine mussels *Mytilis edulis* around Long Island, New York measured the frequency of a particular allele lap^{94} which codes for an enzyme that helps the mussel to regulate its internal saltwater balance.

The data for the above study was presented in the form of the allele frequencies (in percentage) of lap^{94} at 11 different locations. The researchers also noticed that mussel larvae typically swim a long distance away from where they were hatched before they attach themselves to rocks and grow to become adults.

Site	1	2	3	4	5	6	7	8	9	10	11
lap^{94} frequency /%	13	16	16	25	36	37	39	55	59	59	59

A map of Long Island showing the location of the sites and the salinity of the water relative to the distance from the open ocean is shown below.

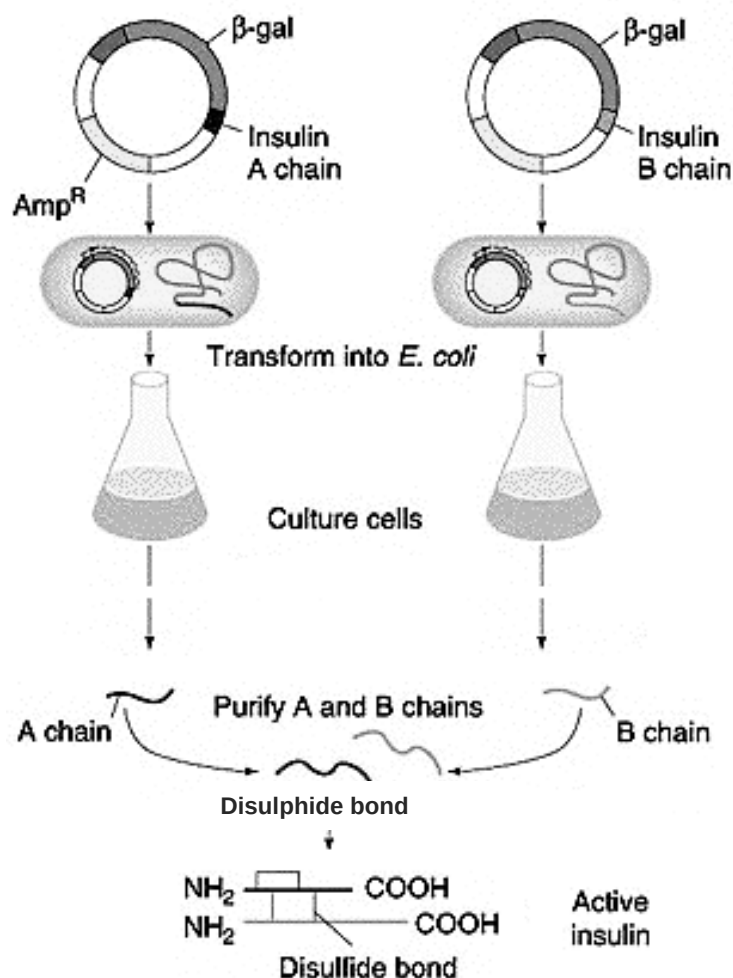


Using the information from the above table, the map and your biological knowledge, which of the statements below regarding the lap^{94} allele is true?

- Selection against the lap^{94} allele is very strong in the southwest portion of Long Island Sound due to low salinity.
- Selection in favour of the lap^{94} allele is very strong in the northeast portion of Long Island Sound due to the high salinity.
- Currently there is still gene flow between the mussels in the different areas which explains the differences in the distribution of the lap^{94} allele frequencies.
- The frequency of the lap^{94} allele will definitely drop to zero in the southwest portion of Long Island in the future.

- A 1 and 2
 B 1, 2 and 3
 C 1, 2 and 4
 D all of the above

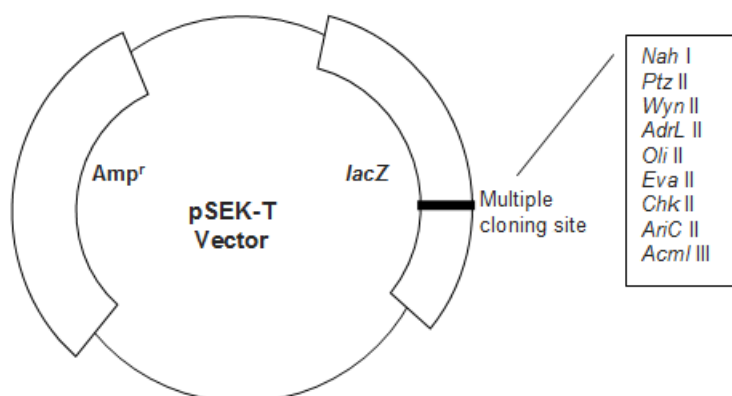
- 35 The insulin hormone is made up of two different polypeptide chains, A-chain and B-chain, joined together by disulphide bonds. The human insulin gene codes for both polypeptide chains. To produce the insulin hormone using *E. coli*, cDNA coding for each polypeptide chain are inserted into separate plasmids and used to transform separate batches of bacteria. The synthesised polypeptide chains are then mixed together to form the hormone.



Which one of the following reasons explains why the two chains have to be produced separately?

- A Bacterial RNA polymerases cannot recognise eukaryotic promoters.
- B Bacterial cells are not able to carry out post-translational modification of insulin.
- C Bacterial cells are unable to excise introns from eukaryotic RNA.
- D Bacterial ribosomes can only translate polycistronic mRNA.

- 36 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the Bt toxin protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for Bt toxin were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	chemicals		colour of colonies
A	ampicillin	X-gal	blue
B	ampicillin	X-gal	white
C	β -galactosidase	X-gal	blue
D	β -galactosidase	lactose	white

- 37 The dashed lines in the template sequence represent a long sequence of bases to be amplified.

Template

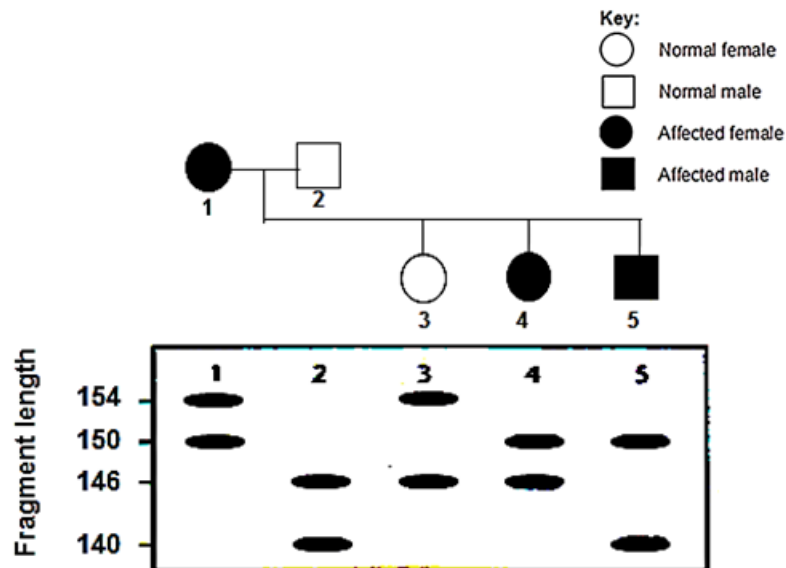
5' ATTCGGA CTTG ----- GTCCAGCTAGAGG 3'

3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

- A 3' TAAGCCT 5' & 5' CTAGAGG 3'
 B 5' ATTCGGA 3' & 3' GATCTCC 5'
 C 3' UAAGCCU 5' & 5' CUAGAGG 3'
 D 5' AUUCGGA 3' & 3' GAUCUCC 5'

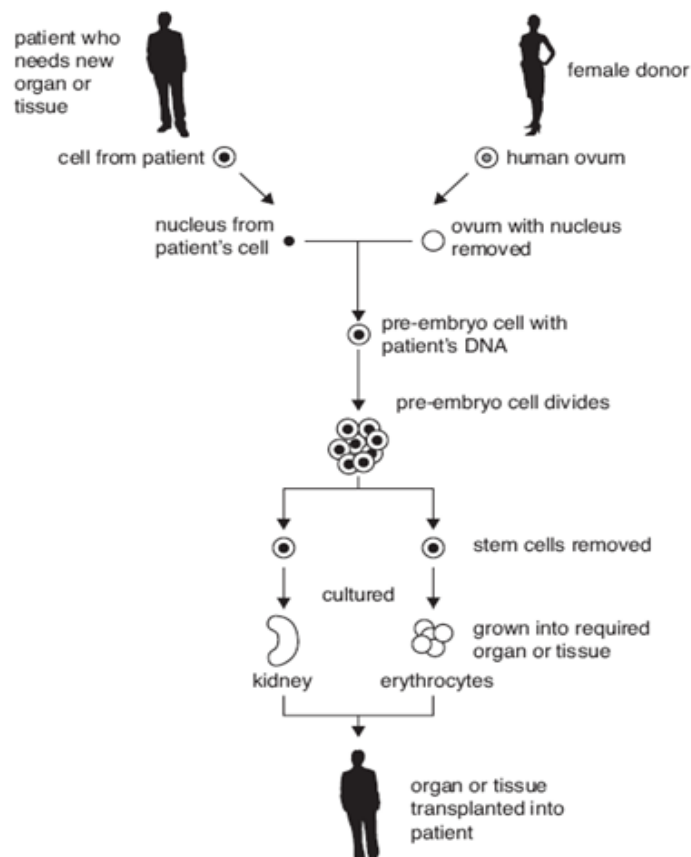
- 38** An RFLP genetic marker located on chromosome 13 is used to trace the inheritance of a genetic disease within an affected family. The following pedigree shows two generations of the family with affected individuals. Results of the Southern Blot analysis for the corresponding family members are shown in the pedigree.



The allelic variants of the polymorphic RFLP locus are designated by the respective fragment lengths produced in this linkage analysis. Which allele of the RFLP locus is linked to the disease allele?

- A** Allele 140
- B** Allele 146
- C** Allele 150
- D** Allele 154

- 39 In the future, therapeutic cloning may produce new tissues and organs for people who are seriously ill. The figure shows stages in producing new tissues and organs by therapeutic cloning.



Which of the following statement about therapeutic cloning is correct?

- A The above procedure can be modified to treat ADA-SCID but not sickle cell anaemia.
 - B The ovum containing the patient's DNA is a diploid cell.
 - C The pre-embryo cells are multipotent cells because they contain an adult patient's DNA.
 - D Therapeutic cloning in this manner is more ethical than using stem cells directly from a donor
- 40 Maize varieties are being developed in which the leaves produce proteins that are toxic to insects. The DNA coding for these toxic proteins was inserted into a maize chromosome via a bacterial plasmid. Many people are opposed to this process.

Which objection is **not** biologically valid?

- A Beneficial insects may be killed if they eat genetically modified maize.
- B Genes for antibiotic resistance are present in plasmids and these genes may be passed to harmful bacteria.
- C Hybridisation may transfer the bacterial genes from maize to weeds, giving the weed species new selective advantage.
- D Mutations may arise in cattle or humans that eat the genetically modified maize.